

# Julie Kallini

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## Education

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|--------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 2023–Present | <b>Ph.D. in Computer Science</b><br>• GPA: 4.26/4.00<br>• Advisors: Christopher Potts and Dan Jurafsky                                   | Stanford University  |
| 2017–2021    | <b>B.S.E. in Computer Science, Minor in Linguistics</b><br>• GPA: 3.94/4.00 ( <i>summa cum laude</i> )<br>• Advisor: Christiane Fellbaum | Princeton University |

## Fellowships, Awards, and Honors

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 2024 | <b>Best Paper Award</b><br>Awarded for <i>Mission: Impossible Language Models</i> , out of over 8,300 submitted papers and 1,700 accepted papers.                                                     | ACL 2024             |
| 2024 | <b>Knight-Hennessy Scholarship Finalist</b>                                                                                                                                                           | Stanford University  |
| 2024 | <b>Hertz Fellowship Semifinalist</b>                                                                                                                                                                  | Stanford University  |
| 2023 | <b>NSF Graduate Research Fellowship</b>                                                                                                                                                               | Stanford University  |
| 2023 | <b>School of Engineering Graduate Fellowship</b>                                                                                                                                                      | Stanford University  |
| 2023 | <b>EDGE Fellowship</b>                                                                                                                                                                                | Stanford University  |
| 2021 | <b>Phi Beta Kappa Honor Society</b><br>Inducted in the top 10% of the graduating class.                                                                                                               | Princeton University |
| 2021 | <b>Phillip Goldman '86 Senior Prize</b><br>Top prize awarded to the graduating senior with the highest academic standing in the computer science class (out of 152 students).                         | Princeton University |
| 2021 | <b>Outstanding CS Senior Thesis Prize</b><br>Awarded for exceptional senior thesis work in computer science (out of ~70 undergraduates who completed a senior thesis).                                | Princeton University |
| 2021 | <b>Outstanding Student Teaching Award</b><br>Awarded for outstanding work as a teaching assistant across several courses in Princeton's computer science department (out of 30 selected TA nominees). | Princeton University |
| 2019 | <b>Tau Beta Pi (Engineering Honor Society)</b><br>Inducted in junior year in the top 12% of engineering students.                                                                                     | Princeton University |

## Teaching Experience

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|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 2026 | <b>Head TA, Natural Language Processing with Deep Learning</b><br>Managed a team of 20 TAs for a class of over 570 students; led weekly TA meetings and held office hours; designed and delivered a new lecture on tokenization and multilinguality. | Stanford University |
| 2024 | <b>Mentor, Introduction to Natural Language Processing</b><br>Designed an introductory natural language processing curriculum and taught a group of 20 high school students daily for 3 weeks.                                                       | AI4ALL              |

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|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 2022–2025 | <b>Instructor, <i>Applied Machine Learning</i></b><br>Built scikit-learn projects and designed course materials for an introductory ML course for working professionals; taught weekly lectures, held office hours, and led one-on-one tutoring sessions. | <i>Uplimit (formerly CoRise)</i> |
| 2021      | <b>Head TA, <i>Machine Translation</i></b><br>Designed new coding exercises and handouts each week on a variety of NLP topics; led students through these collaborative exercises during weekly hour-long practicum classes.                              | <i>Princeton University</i>      |
| 2020      | <b>TA, <i>Computer Networks</i></b><br>Taught course material and ran weekly in-class discussion sessions.                                                                                                                                                | <i>Princeton University</i>      |
| 2018–2019 | <b>Lead Grader/CA, <i>Algorithms and Data Structures</i></b><br>Compiled and edited assignment/exam rubrics alongside instructors; trained and supervised other graders and course assistants during weekly grading meetings.                             | <i>Princeton University</i>      |
| 2018      | <b>Grader/CA, <i>Computer Science: An Interdisciplinary Approach</i></b><br>Attended weekly meetings to grade assignments, projects, and exams; provided constructive feedback on the accuracy and quality of students' Java code.                        | <i>Princeton University</i>      |

## Industry Experience

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|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 2025      | <b>Research Scientist Intern, Tokenization and Model Pre-training</b><br>Designed new training and inference techniques to speed up Meta's Byte Latent Transformer language model, using methods from text diffusion and speculative decoding. | <i>Meta Superintelligence</i> |
| 2021–2023 | <b>Software Engineer, Machine Learning for Ads Privacy</b><br>Built classifiers to detect sensitive topics in media across the Meta family of apps using Transformer-based encoder models. Led data collection efforts.                        | <i>Meta</i>                   |

## Peer-reviewed Publications

- [1] **Julie Kallini**, Artidoro Pagnoni, Tomasz Limisiewicz, Gargi Ghosh, Luke Zettlemoyer, Christopher Potts, Xiaochuang Han, and Srinivasan Iyer. [Fast Byte Latent Transformer](#). In *The Forty-Third International Conference on Machine Learning*, Seoul, South Korea, July 2026.
- [2] Milan Miletić, **Julie Kallini**, and Ekaterina Shutova. [Phonemes to the Rescue: Multilingual Tokenization Based on International Phonetic Alphabet](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, San Diego, California, United States, July 2026.
- [3] **Julie Kallini** and Christopher Potts. [Language Models as Tools for Investigating the Distinction Between Possible and Impossible Natural Languages](#). *Behavioral and Brain Sciences*, 49:e211, July 2026.
- [4] **Julie Kallini**, Dan Jurafsky, Christopher Potts, and Martijn Bartelds. [False Friends Are Not Foes: Investigating Vocabulary Overlap in Multilingual Language Models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Suzhou, China, November 2025.
- [5] **Julie Kallini**, Shikhar Murty, Christopher D. Manning, Christopher Potts, and Róbert Csordás. [MrT5: Dynamic Token Merging for Efficient Byte-level Language Models](#). In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, Tampines, Singapore, April 2025.
- [6] **Julie Kallini**, Isabel Papadimitriou, Richard Futrell, Kyle Mahowald, and Christopher Potts. [Mission: Impossible Language Models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, Bangkok, Thailand, August 2024. **Best Paper Award**.
- [7] **Julie Kallini** and Christiane Fellbaum. [What to Make of make? Sense Distinctions for Light Verbs](#). In *Global WordNet Conference 2023*, Donostia-San Sebastian, Spain, January 2023.
- [8] **Julie Kallini** and Christiane Fellbaum. [Computational Approaches for Understanding Semantic Constraints on Two-termed Coordination Structures](#). In *Text, Speech, and Dialogue*, Brno, Czech Republic, September 2022.

- [9] **Julie Kallini** and Christiane Fellbaum. [A Corpus-based Syntactic Analysis of Two-termed Unlike Coordination](#). In *Findings of the Association for Computational Linguistics: EMNLP 2021*, Punta Cana, Dominican Republic, November 2021.

## Invited Talks

|          |                                                                                |                                                    |
|----------|--------------------------------------------------------------------------------|----------------------------------------------------|
| Feb 2026 | <a href="#">Tokenization and Multilinguality</a>                               | Stanford CS224N Guest Lecture Series               |
| Oct 2025 | <a href="#">A Case for Tokenizer-free Language Models</a>                      | Stanford NLP 25th Alumni Reunion                   |
| May 2025 | <a href="#">Dynamic Token Merging for Efficient Byte-level Language Models</a> | UCLA NLP Seminar                                   |
| Mar 2025 | <a href="#">Dynamic Token Merging for Efficient Byte-level Language Models</a> | USC ISI NLP Seminar                                |
| Nov 2024 | <a href="#">Mission: Impossible Language Models</a>                            | UCLA Coalas Lab                                    |
| Oct 2024 | <a href="#">Mission: Impossible Language Models</a>                            | USC ISI NLP Seminar                                |
| Oct 2024 | <a href="#">Mission: Impossible Language Models</a>                            | UChicago/TTIC Natural Language Processing Seminar  |
| Sep 2024 | <a href="#">Mission: Impossible Language Models</a>                            | UT Austin Computational Linguistics Research Group |
| Apr 2024 | <a href="#">Mission: Impossible Language Models</a>                            | MIT Computational Psycholinguistics Lab            |
| Apr 2024 | <a href="#">Mission: Impossible Language Models</a>                            | Bay Area Language Processing Interest Group        |
| Mar 2024 | <a href="#">Mission: Impossible Language Models</a>                            | NLP-GenAI Online Seminar, Boğaziçi University      |

## Press Coverage

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|----------|------------------------------------------------------------------------------------------|------------------|
| Apr 2025 | <a href="#">When ChatGPT Broke an Entire Field: An Oral History</a>                      | Quanta Magazine  |
| Mar 2025 | <a href="#">Dynamic Token Merging for Efficient Byte-level Language Models</a>           | TWIML AI Podcast |
| Jan 2025 | <a href="#">Can AI Models Show Us How People Learn? Impossible Languages Point a Way</a> | Quanta Magazine  |

## Volunteering and Outreach

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|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 2022–2024 | <b>AI Mentor</b><br>Mentored students from underrepresented backgrounds to pursue careers in data science, ML, and AI, including job application and interview preparation. | Break Through Tech AI           |
| 2021–2023 | <b>Alumni Interview Volunteer</b><br>Conducted interviews with high school students applying to Princeton.                                                                  | Princeton University Admissions |

## Academic Service

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|------|----------------------------|------------------------------------------------------------------|
| 2026 | <b>Workshop Reviewer</b>   | Tokenization Workshop (COLM 2026)                                |
| 2026 | <b>Conference Reviewer</b> | International Conference on Machine Learning (ICML 2026)         |
| 2026 | <b>Conference Reviewer</b> | ACL Rolling Review (ACL 2026 Cycle)                              |
| 2025 | <b>Conference Reviewer</b> | ACL Rolling Review (EMNLP 2025 Cycle)                            |
| 2025 | <b>Conference Reviewer</b> | International Conference on Learning Representations (ICLR 2025) |
| 2025 | <b>Workshop Reviewer</b>   | Tokenization Workshop (ICML 2025)                                |
| 2024 | <b>Workshop Reviewer</b>   | System-2 Reasoning at Scale Workshop (NeurIPS 2024)              |